Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The UDP protocol reveals that: The computer sent DNS queries using the UDP protocol to resolve the IP address of www.yummyrecipesforme.com. However, the requests failed, and the computer received ICMP error messages. **This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message:** The ICMP error message stated "UDP port 53 unreachable." This indicates that the DNS queries could not reach an available service on port 53. The ICMP message was an error response, not a regular ICMP echo reply. **The port noted in the error message is used for:** Port 53 is commonly used by the Domain Name System (DNS), which translates domain names into IP addresses. **The most likely issue is:** The DNS service was unavailable or not listening on port 53 on the destination DNS server. As a result, the computer could not resolve the website's domain name into an IP address and could not access the website.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
Time incident occurred: The incident was observed at approximately 13:24:32, as indicated by the timestamps in the tcpdump log. The first recorded query occurred at 13:24:32.192571. **Explain how the IT team became aware of the incident:** Several customers reported that they could not access www.yummyrecipesforme.com and received a "destination port unreachable" error after waiting for the website to load. The cybersecurity analyst attempted to access the website and experienced the same problem. **Explain the actions taken by the IT department to investigate the incident:** The cybersecurity analyst used the tcpdump network protocol analyzer to capture and inspect network traffic while attempting to load the website. The analyst examined the source and destination IP addresses, timestamps, protocols, port numbers, and error messages to

identify the cause of the connectivity problem.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): The tcpdump log showed that the computer, with source IP address 192.51.100.15, sent DNS queries using UDP to the DNS server at 203.0.113.2. The queries were directed to port 53, which is commonly used by DNS. The DNS queries failed, and ICMP error messages indicated that UDP port 53 was unreachable. The same error occurred repeatedly when the requests were retried. This prevented the computer from resolving the domain name to an IP address and accessing the website.

Note a likely cause of the incident: A likely cause is that the DNS service on the destination server was down, misconfigured, or not listening on port 53. A firewall or network configuration issue could also have blocked DNS traffic. Further investigation is needed to determine the exact root cause. The IT team should verify the DNS service status, check firewall rules and network configurations, and confirm that the server is listening on port 53.